

Methods for using High-Resolution Measurements to Identify Effects of Aquatic Vegetation on Water-Quality and Vertical Stratification

Water quality was measured on July 13 and October 10, 2018 in FI, and July 14 and October 11 2018 in LHT (Figure 1) using a combination of in situ, high resolution measurements and discrete water sample collection and subsequent analysis. The timing of each survey was chosen in order to highlight the combination of spring/neap and solar diel cycles. The July and October surveys were conducted at the same point in the spring-neap cycle near the spring tide when stage is highest. Both surveys were conducted when the low tide coincided with solar noon. The surveys took place at slightly different phases of the semidiurnal tide with the July maps following the high-high tide and during the low-low tide of the day and the October survey following the low-high tide and during the high-low tide. Sample plans were determined by semidurnal and diurnal tides; the Tidal Trend post-processing tool described below was used to place water quality parameters in the context of the semidiurnal and diurnal tidal cycle.

The in situ data was collected using a boat based high-resolution mapping survey system developed by the USGS CAWSC (Downing et al., 2016; Bergamaschi et al. 2020a, b). All data were collected in Pacific Standard Time (PST). The set up was modified to include the addition of an aluminum boom on the bow of the boat to minimize disturbance of the water being sampled, and to allow vertical profile water-quality measurement to be made in the aquatic vegetation beds. In addition, the boat was equipped with spuds that could be lowered into the sediment to keep the boat in place when measurements were being made at a specific location. The methodology is described in greater detail below.

Site Description

Both FI and LHT are flooded agriculture islands within the Cache Slough Complex (CSC), a region where many wetland restoration efforts in support of native fish habitat are ongoing or planned (California EcoRestore, 2020). The CSC is downstream of wastewater treatment effluent and local agricultural inputs high in dissolved inorganic nitrogen (DIN). Under low flow conditions when surface run-off is minimal and dam releases upstream are reduced, DIN is dispersed from south to north across the region by tides. LHT is a ~570,000 m² (~140 acre) site at the southern extent of the CSC, adjacent and hydrologically connected to Cache Slough and has higher ambient DIN concentrations relative to FI, a slightly smaller tract (~525,000 m², ~130-acres) that is hydrologically connected to Shag Slough (Figure 1). The levees surrounding FI are mostly intact; water predominantly enters this area through an opening to Shag Slough to the east and an opening to Cache Slough to the northwest. In notable contrast, the levees around LHT are more fragmented, particularly to the south where water can easily exchange with water entering this region from the south; LHT also has two larger breaches to the southeast, northwest, and north. During the study period both LTH and FI supported dense populations of submerged aquatic vegetation (SAV) beds; particularly dominant species included Brazilian Waterweed (*Egeria densa*) and Coontail, (*Ceratophyllum demersum*). Populations of floating aquatic

vegetation (FAV), notably Water Hyacinth (*Eichhornia crassipes*) and Water Primrose (*Ludwigia spp*) were present at some margins, particularly the northern area of FI.

Although attempts were made to take visual notes on the presence and density of aquatic vegetation during field activities, the ability to see into the water column varied greatly. For example, during lower tides and low winds it was easy to see into the water column and take notes and photographs of the SAV, but during higher tidal stages and/or higher winds the same areas would appear free of SAV simply because one could not see into the water column. During some periods, particularly in the afternoon in July when temperatures were high, green algal mats covered the SAV beds and floated to the surface (Figure 2).



Figure 1. Little Hastings Tract and French Island study locations in the Cache Slough Complex.

Sampling and analysis

High-resolution water sampling: Mapping and profiles

Data collection was conducted using the USGS R/V Mary Landsteiner, which was outfitted with an aluminum boom (~ 7 meters long) that mounted on the bow of the research vessel and extended up to 5 meters in front of the bow (Figure 2). A high resolution 16 Hz CTD with an internal pump (FastCat, Sea-Bird Scientific, Bellevue, WA) was attached to the end of the boom, and along with a second pump on deck routed water from the CTD to the flow-through system via Tygon tubing; sample tubing and deployment line, both attached to the CTD, were laid over the sample boom. Sample water was continuously pumped from the CTD onto the boat while underway at low speed <5 mph (2 m/s) or when stationary. To avoid clogging, the intake to the CTD was covered with a coarse 40 mesh screen (0.42 sq. mm, 400 micron). CTD data was collected and monitored in real-time to allow for some control over sample depth. Underwater cameras (usb endoscopes) equipped with LED lights were also fastened to the CTD in order to observe in real time when the instrument approached the sediment surface; this helped ensure a full water profile was collected and helped avoid hitting the sediment surface and clogging the flow-through sampling system. However, in some cases dense aquatic vegetation made it challenging to see, resulting in cases when the bottom was disturbed and other cases when the CTD was not lowered sufficiently to get a complete profile.



Figure 2. Photos showing how the CTD and tubing were mounted on a boom that extended ~6 meters in front of the boat, allowing for minimal disturbance of the water column during mapping. Photos also highlight how wind and wave activity, as well as water stage related to tidal variability, controlled whether SAV was visible in the water column. Bottom right photo shows the dense green algal mats that were sometimes present on the surface.

Sample water delivered from the CTD first passed through a 178 μm in-line strainer (Cole Parmer; EW-29595-47, Vernon Hills, IL) to remove large debris before splitting the flow into separate analytical paths to water-quality instrumentation. Sample water was split into the following flow paths: (1) A flow-through system consisting of a thermosalinograph that recorded temperature, conductance, and salinity (Sea-Bird Scientific SB45 (TSG), Bellevue, WA); fluorometers that measured fluorescence of chlorophyll (fCHL) and fluorescence of dissolved organic matter (fDOM, WETLabs WETstar (WS), Philomath, OR); a beam transmissometer that recorded transmittance and attenuation (WETLabs model C-Star transmissometer (CStar), Philomath, OR), a nitrate (NO_3) analyzer (SUNA V2; Sea-Bird Scientific, Bellevue, WA). (2) A flow chamber for a multiparameter water quality sonde (YSI EXO2; Xylem Inc. (EXO), Rye Brook, New York) equipped with sensors to measure temperature, specific conductance

(SpC), turbidity, pH, dissolved oxygen (DO), fDOM and fCHL that then passed water to a fluorometer designed to measure different algal classes (FluoroProbe III (FP); BBE Moldaenke, Kiel, Germany). (3) An open-split interface at atmospheric pressure that served water filtered through a 0.45 micron high-capacity in-line groundwater sampling capsule filter (PALL Corporation, Port Washington, New York) to the on-board ammonium (NH_4) analyzer. The Fluoroprobe was only used for the October transects.

The NH_4 analyzer was a continuous flow, gas diffusion/conductivity-based (Carlson 1986) analyzer for ammonium (NH_4) analysis (TL-2800; Timberline Instruments, Boulder, CO) that was modified for field operation and continuous data collection by the manufacturer. Modifications included installation in a ruggedized housing, addition of an automated line-switching valve, addition of a heating unit to maintain the instrument at a constant above-ambient temperature, and changes to the software. Full standard curves were run at the beginning and end of each day and partial curves and blank water were run throughout the day each time the boat stopped to sample. The NH_4 analyzer was connected to a stand-alone computer and collected data through its native software. Due to technical difficulties, the ammonium data for July transects did not pass quality assurance and is not included in this report.

All instrumentation was cleaned, and calibrations were checked prior to each use following the manufacturer's recommendation or as described below. The CTD data were recorded at a rate of ~16 measurements per second on a separate computer to accurately record depth. Instruments on the first and second flow path were recorded at 1-second frequency on a single data logger (CR6, Campbell Scientific, Logan, Utah, United States) together with a timestamp and boat position obtained from a high-resolution global positioning system receiver (16X-HVS, Garmin, Olathe, Kansas). The FluoroProbe logged data internally and to the host software. All data were displayed in real time so scientists on board could respond when they observed changes relevant to study objectives, or if issues with flow or instrument performance were noted (e.g., high turbidity or bubbles which interfere with optical measurements and can clog the tubing and filters).

High-resolution measurements at low speed or stationery – also referred to as high-resolution mapping - were completed across both LHT and FI on individual days from about 8 am to 5 pm. Mapping data were collected at a targeted sample depth between 0.1 – 0.6 m from the surface following a transect in the shape of a figure 8 in order to take measurements across the tracts at different points in the diurnal and tidal cycle (Figure 3). At least two figure-8 transects were collected each day. Vertical profiles were completed at six or seven pre-designated stations along each figure 8-transect, and replicate profiles were periodically collected. For vertical profiles, the boat was anchored by steel shafts – spuds – that were lowered from the boat into the sediment below to prevent lateral movement during sampling. Vertical profile measurements extended up to about 2.0 m depth in LHT and FI, depending on the tide (Table 1). Vertical profiles were also occasionally taken in the channels at the eastern entrance to the tracts (Shag Slough for FI and Cache Slough for LHT).

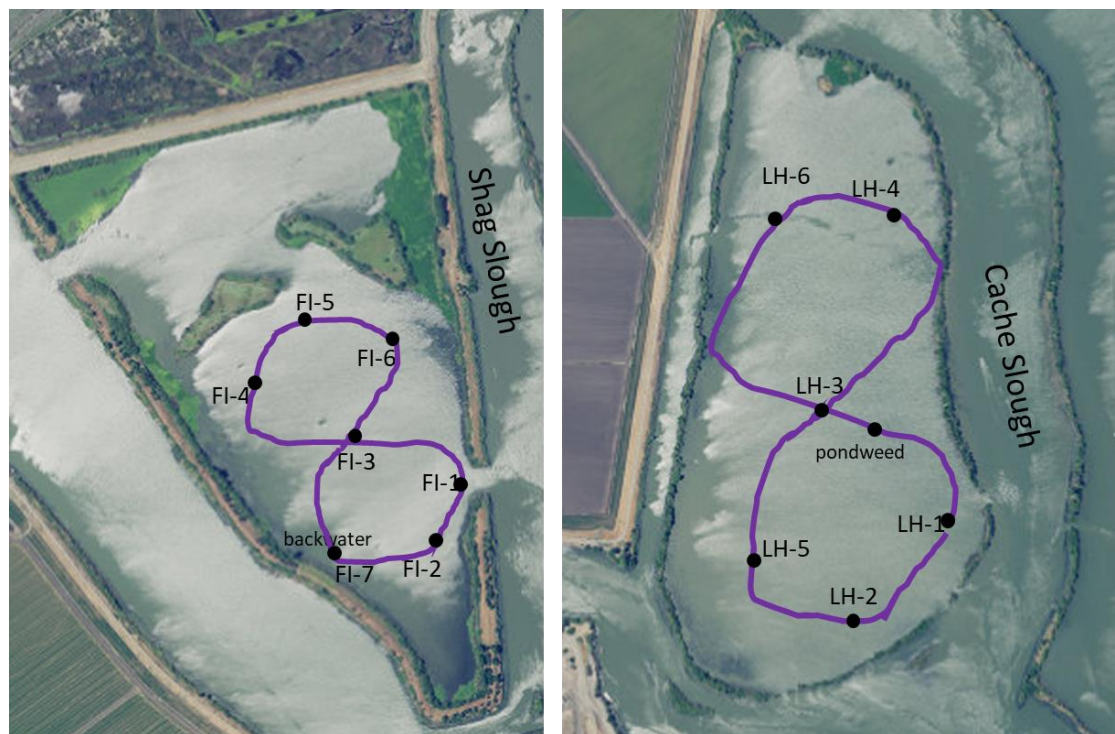


Figure 3. Station locations for profiles on French Island (FI, left) and Little Hastings Tract (LHT, right) indicated in black circles, with purple lines showing the approximate track of the figure eight transects.

French Island - 7/13/18				French Island - 10/10/18		
time	8:30 - 11:20	11:45 - 13:40	14:15 - 16:10	8:58 - 11:44	11:45 - 14:14	14:15 - 15:58
PROFILE	Figure 8-1 depth (m)	Figure 8-2 depth (m)	Figure 8-3 depth (m)	Figure 8-1 depth (m)	Figure 8-2 depth (m)	Figure 8-3 depth (m)
Shag Slough	--	--	--	3.5	--	--
FI - 1	2.5	--	--	--	1.7	2.4
FI - 2	1.6	--	--	1.2	0.7	1.7, 1.4
FI - 3	1.5	1.4	1.1	1.2	0.9	1.3
FI - 4	1.5	1	1.2	0.9	0.6	--
FI - 5	1.4	--	1.8	1.7	1.1	--
FI - 6	1.4	0.9	1.5	0.7	1.4	--
FI-3	1	0.8	0.9	1	1.5	--
FI-backwater	1	--	--	--	--	--
FI-7	1	--	--	1.09	1.4	--
FI-1	1.5	--	1.2	--	--	--
FI-2	0.9	0.9	--	--	--	--
Little Hastings Tract- 7/14/18				Little Hasting Tract- 10/11/18		
time	10:45 - 13:56	13:57 - 16:09	--	8:59 - 12:18	12:19 - 15:35	--
PROFILE	Figure 8-1 depth (m)	Figure 8-2 depth (m)	--	Figure 8-1 depth (m)	Figure 8-2 depth (m)	--
Cache Slough	--	--	--	3.5	3.7	--
LH-2	1.1	--	--	1.7, 1.3	1	--
pondweed	--	--	--	0.9	--	--
LH-3	1	1	--	1.3	--	--
LH-6	--	0.85	--	0.8	--	--
LH-4	0.9	0.9	--	1.05	--	--
LH-3	0.7	1.2	--	0.9	--	--
LH-5	0.5	1.4	--	--	1.1	--
LH-6	--	--	--	--	1.2, 1.1	--
LH-5	--	--	--	--	1.1	--
LH-2	0.8	1.5	--	0.9	1.6, 1.6	--
LH-3	--	--	--	--	--	--
LH-1	--	--	--	0.9	1.3	--

Table 1. Summary of vertical profiles taken at each station (e.g., FI-1, FI-2, FI-3....) during the different figure eight transects (Figure 8-1, Figure 8-2...) that were taken throughout each day for French Island (FI; 7/13/18, 10/10/18) and Little Hastings Tract (LHT; 7/14/18, 10/11/18). Periodically, replicate profiles were taken (denoted by ‘,’ between depth).

Discrete water sampling

Approximately 15 to 25 discrete water samples for laboratory analysis were collected each day at designated sampling stations using a fourth flow path. Samples were collected near the surface or at a depth approximately 0.2 m above the bed sediment. Unfiltered water samples for analysis of chlorophyll-a (Chl-a), pheophytin, phycocyanin, allophycocyanin, and phycoerythrin were pumped directly into 1 L brown HDPE bottles. Filtered discrete water samples were passed through an inline

0.45-micron filter for determination of ammonium (NH_4), nitrate plus nitrite ($\text{NO}_3 + \text{NO}_2$), nitrite (NO_2), organophosphate (PO_4), total dissolved nitrogen (TDN), dissolved organic carbon (DOC), and optical properties of dissolved organic matter (absorbance and fluorescence). Samples for DOC and optical measurements were collected into baked 125 mL amber glass bottles and water for the other analyses were collected into brown 125 mL HDPE bottles. Samples for analysis of water isotopes – oxygen ($\delta^{18}\text{O}$) and hydrogen ($\delta^2\text{H}$) – were collected into 20 mL vials free of headspace.

All samples were stored on ice in the dark during transit to the laboratory where they were stored at 2°C . Dissolved organic carbon samples were preserved to <2 pH with high-purity sulfuric acid the day following collection and stored at 2°C . Samples for Chl-a analysis were filtered within 24 hours of collection through a $0.3\ \mu\text{m}$ nominal pore size glass fiber filter (Advantec MFS, Inc, Dublin, CA). Frozen Chl-a samples and chilled nutrient samples were shipped to the USGS National Water Quality Laboratory (Denver, CO) for analysis. Samples for DOC and optical properties were delivered to the USGS Organic Matter and Research Lab (OMRL, Sacramento, CA).

Laboratory analyses

Concentrations of nutrients were determined colorimetrically using an automated-segmented flow analyzer. Concentrations of nitrogen as nitrite ($\text{NO}_2\text{-N}$) and as nitrate plus nitrite ($\text{NO}_3\text{-N} + \text{NO}_2\text{-N}$) were determined by colorimetric analysis (Fishman, 1993; Patton and Kryskalla, 2011), ammonium as nitrogen ($\text{NH}_4\text{-N}$) was determined by colorimetric analysis after reaction with salicylate-hypochlorite (Fishman, 1993), and total dissolved nitrogen (TDN) was determined by alkaline persulfate digestion (Patton and Kryskalla 2003). Orthophosphate as phosphorous ($\text{PO}_4\text{-P}$; also referred to as soluble reactive phosphate) was determined by colorimetric analysis after reaction with NH_4 molybdate and reduction with ascorbic acid (Patton and Truitt, 1992). Chlorophyll-a and phaeophytin concentrations were determined according to EPA Method 445.0 (Arar and Collins, 1997) using sonication extraction. Dissolved organic carbon concentrations were measured using a total organic carbon analyzer (TOC- V_{CSH} , Shimadzu Scientific Instruments, Columbia, Maryland) using high-temperature catalytic combustion according to a modified version of USEPA method 415.3 (Potter and Wimsatt, 2005). Optical properties of absorbance spectra and fluorescence matrices were measured in an acid-cleaned 1 cm quartz cuvette using a spectrophotometer equipped with a CCD detector (Aqualog®; Horiba Instruments, New Jersey, U.S.A.) (Hansen et al., 2016). Nutrient concentrations at the method detection limit (MDL) were assigned a value that was half of the MDL. The MDL half values assigned to $\text{NH}_4\text{-N}$, $\text{NO}_3\text{-N} + \text{NO}_2$, and $\text{NO}_2\text{-N}$, are $0.36\ \mu\text{M}$, $1.43\ \mu\text{M}$, and $0.04\ \mu\text{M}$, respectively.

Data processing

Data from onboard continuous instruments not directly logged to the flow-through data collection system (e.g. Timberline, CTD, and Fluoroprobe) were merged based on timestamp to the nearest second. To account for changes in flow rates that may occur when sediment and algae clogged the CTD intake mesh filter, a lag tool designed in an Excel macro was used to more effectively merge data using observational data described below. Given our best efforts, the methods described here are part of a pilot study and it should be noted that some offset or lag in data may still exist. The CTD data required conversion from ASCII, a median filter to bin data from 16 Hz to 1 Hz, and a time lag correction applied –

determined by comparing the CTD temperature and specific conductivity data to the on-board instruments - prior to the timestamp merge. All data directly logged to the flow-through data collection system was processed using pandas software library (The pandas development team, 2020) in Jupyter Notebook (Project Jupyter) to remove periods of compromised data (e.g. flow blockages, bubbles), to apply instrument corrections and unit conversions. It was determined that temperature data reported by the CTD was sometimes measuring air temperatures rather than water temperature, thus these data are not reported. The EXO and thermosalinograph temperature data had the strongest relationship with the CTD depth data and are therefore the primary data streams for this parameter. Data offsets and correction are documented in the associated ScienceBase release (<https://www.sciencebase.gov/catalog/item/5f1f44c582cef313ed8ec753>).

Final NO_3 values were obtained by regressing nitrate concentration data reported by the in situ instrument against NO_3 concentrations obtained from laboratory measurements for the July and October field efforts. Individual sample pairs that were more than three standard deviations from the initial regression were judged to be outliers and removed from the regression, and the regression rerun. The high-resolution nitrate data was corrected with a regression for July ($R^2 = 0.87$; $y = 0.8645x - 0.3967$) and October ($R^2 = 0.95$; $y = 0.9974x - 4.6161$) with regression equations identified in the data dictionary and

NH_4 data was processed for the October sampling dates by first correcting the voltage output of the instrument for baseline drift based on the instrument response during periodic measurements of organic-free deionized water ($n \geq 10$) and then using a regression model of the voltage response to the standards run that day. A regression coefficient of $>0.97 R^2$ or better was judged to be acceptable. The NH_4 data timestamp was corrected to account for the time lag associated with the microfluidic processing prior to integration with other data sources.

Estimated high resolution (1 sec) dissolved organic carbon (estDOC) values were obtained by regressing laboratory reported DOC concentrations against high-resolution fDOM (QSU) values reported by the WetStar using the same outlier detection methodology as NO_3 values. The estimated DOC values were obtained with a regression for July ($R^2 = 0.57$; $y = 0.0697x + 0.6453$) and October ($R^2 = 0.48$; $y = 0.0684x + 0.4583$)

Mapping and profile data were examined using data visualization software (Tableau 2020.1.3) that allowed us to review data both temporally and spatially. Data was “coded” (metadata was assigned) so that parameters could be easily associated with either surface water quality mapping data collected in association with the figure eight transects when the CTD was between $\sim 0.1 - 0.6$ m below the surface (“mapping”) or with the depth profiles that were taken while stationary at specific sites (“profile”). Data that were associated with both surface water (0.2-0.6 m) and vertical profiles were designated as “mapping + profile”; this ensured both data types were more complete and also allowed us to associated discrete data with both surface mapping and profile data sets.

A centered 20 second median filter was applied to each parameter in the mapping time series data (except for the Timberline NH_4 data) to assist with outlier removal. No filter was applied to data

associated with the downcast and upcast for the depth profiles. Further quality assurance and data validation was completed to remove data compromised by sediment or bubbles in the sample lines.

Dissolved inorganic nitrogen (DIN) concentration was calculated as the sum of NO_3 , NO_2 and NH_4 —both for the continuous data and the discrete sample data. High resolution total dissolved nitrogen (TDN) was calculated as the sum of DIN and DON.

In the comparison of discrete sample data to high resolution data there were three extracted chlorophyll-a laboratory samples that were not approved as concentrations were improbably high relative to in-situ optical measurements. These outliers were assumed to reflect disturbance of particles in submerged aquatic vegetation beds or bottom sediment.

TidalTrends Post Processing tool

To place high-resolution water quality data into the context of tides, the TidalTrend software program (soon to be open source) used velocity data collected at the U.S. Geological Survey's Cache Slough at Ryer Island continuous monitoring station. TidalTrend uses stage or velocity data – reported at 15 minute intervals - to parameterize semidiurnal (high-low cycles) and diurnal tidal limbs (spring-neap cycles) and is further described by Colombano et al. (2020). Briefly, when the 15 minute TidalTrend data is merged with the high-resolution water quality data, the normalized hydrologic data allows for water quality and other ecological data to be grouped and interpreted without the danger of tidal aliasing. TidalTrend was applied to velocity data at Cache Slough at Ryer Island (USGS Station 11455350, https://waterdata.usgs.gov/ca/nwis/uv?site_no=11455350) because the dataset was more complete relative to the station closer to the flooded islands (Cache Slough at Liberty Island 11455315, https://waterdata.usgs.gov/ca/nwis/uv?site_no=11455315). In an earlier exercise, we determined stage at the flooded islands was in phase with the Liberty Island stage data – for the time period of this study, we determined that velocity at Liberty and Ryer stations were within 15-30 minutes of each other (data not shown).

The TidalTrend tool parameterized velocity data at Cache Slough at Ryer Island over the period between January 1, 2018 – April 1, 2019 to best capture the hydrological variability within the system and better parameterize the diurnal and semidiurnal signals. The following parameters are described in the metadata accompanying the data release

(<https://www.sciencebase.gov/catalog/item/5f1f44c582cef313ed8ec753>): normalized input signal, diurnal range, and normalized diurnal range. The 'normalized input signal,' that is mean water velocity in this example, is the input signal normalized where the semidiurnal low is 0 and the semidiurnal high is 1. 'Diurnal range' is the absolute difference between the 'diurnal low' (lowest period in the 24.8-hour tidal period) and the 'diurnal high' (highest value over a 24.8-hour tidal period). The 'normalized diurnal range' normalizes the diurnal range so the 14.8 day low is 0 and the 14.8 day high is 1, representing the neap and spring tides, respectively.

References

- Arar, E.J. and Collins, G.B. (1997) Method 445.0: In vitro determination of chlorophyll a and pheophytin a in marine and freshwater algae by fluorescence. United States Environmental Protection Agency, Office of Research and Development, National Exposure Research Laboratory.
- Bergamaschi, B.A., Kraus, T.E., Downing, B.D., Soto Perez, J., O'Donnell, K., Hansen, J.A., Hansen, A.M., Gelber, A.D., and Stumpner, E.B., 2020. Assessing spatial variability of nutrients and related water quality constituents in the California Sacramento-San Joaquin Delta at the landscape scale: High resolution mapping surveys: U.S. Geological Survey data release, <https://doi.org/10.5066/P9FQEUAL>.
- Bergamaschi, B.A., Kraus, T.E., Downing, B.D., Stumpner, E.B., O'Donnell, K., Hansen, J.A., Soto Perez, J., Hansen, A.M., Gelber, A.D., in review. Assessing spatial variability of nutrients and related water quality constituents in the California Sacramento-San Joaquin Delta at the landscape scale: 2018 High resolution mapping surveys. U.S. Geological Survey Scientific Investigations Report.
- Bennett, A. and Bogorad, L. (1973) Complementary chromatic adaptation in a filamentous blue-green alga. *The Journal of cell biology* 58, 419-435.
- Colombano, D.D., Donovan, J.M., Ayers, D.E. et al. Tidal effects on marsh habitat use by three fishes in the San Francisco Estuary. *Environ Biol Fish* 103, 605–623 (2020). <https://doi.org/10.1007/s10641-020-00973-w>
- California EcoRestore, 2020. <https://water.ca.gov/Programs/All-Programs/EcoRestore>. California Department of Water Resources. Assessed September 2, 2020.
- Carlson, R.M. (1986). Continuous flow of reduction of nitrate to ammonia with granular zinc. *Analytical Chemistry* 58, 7, 1590-1591.
- Downing, B.D., Pellerin, B.A., Bergamaschi, B.A., Saraceno, J.F. and Kraus, T.E., 2012. Seeing the light: The effects of particles, dissolved materials, and temperature on in situ measurements of DOM fluorescence in rivers and streams. *Limnology and Oceanography: Methods*, 10(10), pp.767-775.
- Donovan, J. & Ayers, D. (2019). TidalTrend software technical report, US Geological Survey Dept of Interior Downing, B.D., Bergamaschi, B.A., Kendall, C., Kraus, T.E.C., Dennis, K.J., Carter, J.A. and Von Dessonneck, T.S. (2016) Using Continuous Underway Isotope Measurements to Map Water Residence Time in Hydrodynamically Complex Tidal Environments. *Environmental Science & Technology* 50, 13387-13396.
- Fishman, M.J. (1993) Methods of analysis by the US Geological Survey National Water Quality Laboratory: Determination of inorganic and organic constituents in water and fluvial sediments. US Geological Survey open-file report (USA).
- Grenier, L., Grossinger, R. (2016) Tools for landscape-scale restoration planning in the Delta. Presentation to the Delta Independent Science Board, January 16, 2013, 2nd Floor Conference Center, Sacramento, CA.
- Hansen, A. M., Kraus, T. E., Pellerin, B. A., Fleck, J. A., Downing, B. D., & Bergamaschi, B. A. (2016). Optical properties of dissolved organic matter (DOM): Effects of biological and photolytic degradation. *Limnology and Oceanography*, 61(3), 1015-1032.
- The pandas development team, 2020. pandas-vel/pandas: pandas. Zenodo v. 1.1.3 <https://doi.org/10.5281/zenodo.3509134>
- Patton, C.J. and Kryskalla, J.R. (2003) Methods of analysis by the U.S. Geological Survey National Water Quality Laboratory: Evaluation of Alkaline Persulfate Digestion as an Alternative to Kjeldahl

- Digestion for Determination of Total and Dissolved Nitrogen and Phosphorus in Water. US Geological Survey. Water-Resources Investigations Report 03-4174.
- Patton, C.J. and Kryskalla, J.R. (2011) Colorimetric determination of nitrate plus nitrite in water by enzymatic reduction, automated discrete analyzer methods. US Geological Survey Techniques and Methods 34.
- Patton, C.J. and Truitt, E.P. (1992) Methods of Analysis by the US Geological Survey National Water Quality Laboratory: Determination of the Total Phosphorus by a Kjeldahl Digestion Method and an Automated Colorimetric Finish that Includes Dialysis. US Geological Survey.
- Potter, B. and Wimsatt, J. (2005) Determination of total organic carbon and specific UV absorbance at 254 nm in source water and drinking water. EPA Document: EPA.
- Rasmussen, N., L. Conrad, H. Green, S. Khanna, J. Caudill, P. Gilbert, P. Goertler, H. Wright, K. Hoffmann, S. Lesmeister, J. Jenkins, L. Takata, D. Bosworth, T. Flynn, E. Hard, T. Sommer. 2020. 2017-2018 Delta Smelt Resiliency Strategy Action for Enhanced Control of Aquatic Weeds and Understanding Effects of Herbicide Treatment on Habitat. Interagency Ecological Program Technical Report. 324 pp.
- Saraceno, J.F., Shanley, J.B., Downing, B.D. and Pellerin, B.A., 2017. Clearing the waters: Evaluating the need for site-specific field fluorescence corrections based on turbidity measurements. *Limnology and Oceanography: Methods*, 15(4), pp.408-416.
- Terzopoulos, D. and Witkin, A. (1988) Physically based models with rigid and deformable components. *IEEE Computer Graphics and Applications* 8, 41-51.
- U.S. Geological Survey, 2017. Metadata Wizard version 2.06. U.S. Geological Survey Fort Collins Science Center. Project page: <https://github.com/usgs/fort-pymdwizard>
- Work, P.A., Downing-Kunz, M. and Drexler, J.Z., 2020. Trapping of Suspended Sediment by Submerged Aquatic Vegetation in a Tidal Freshwater Region: Field Observations and Long-Term Trends. *Estuaries and Coasts*, pp.1-16.